Exploring the Best CFA Model for Balanced Likert Scales

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Abstract

Reverse worded(RW) items are written so that agreement indicates a low endorsement of the construct. RW items are designed to cancel out acquiescence bias (Ray, 1983) when the item scores are averaged across the items. However, RW items may affect the covariance structure and disrupt the intended factor structure, introducing extraneous factors (Carmines & Zeller, 1979; Greenberger et al., 2003). To address these challenges, multiple new factor models have been proposed to better account for the influence of RW items on Likert scales and have yielded a much better fit than the theoretical factor model across several different Likert scales (e.g., Argyle et al., 1989; Lindwall et al., 2012; Savalei & Falk, 2014). However, it is unclear which of these new models can consistently yield the best fit across different Likert scales, especially given the differing degrees of freedom and complex nesting structures among these models. Our present research aims to investigate the most suitable model for Likert scales with RW items. Specifically, we investigated the nesting structures among these models using a statistical procedure called the nesting and equivalence testing (NET) procedure (Bentler & Satorra, 2010) and then using real data for multiple Likert scales, we compared the nested models using a fit index for comparing nested model called RMSEA_D (Savalei et al., 2023). Our results showed that the model fit was similar for the more parsimonious CTCM models, CTCU models, and CTCM-PW/RW models have a substantial improvement in model fit compared with the other CTCM models, suggesting that CTCM-PW/RW should be the preferred model considering model simplicity and efficiency.

Keywords: structural equation modeling, nested models, root mean square error of approximation, chi-square difference tests

Exploring the Best CFA Model for Balanced Likert Scales

The Likert format scale is among the most popular response scales in the social sciences, where the question stem typically prompts participants to express their level of agreement with a statement, and the response options usually follow a format ranging from "strongly disagree" to "strongly agree" (Baumgartner & Steenkamp, 2001). Typically, Likert scales consist of an equal number of positively worded (PW) and reverse worded (RW) items. PW items are framed to align with the measured construct (e.g., the item, *I enjoy socializing*, on a scale measuring extroversion), whereas RW items are framed in the opposite direction of the construct being assessed (e.g., the item, *I prefer solitude*, on a scale measuring extroversion) (e.g., Greenleaf, 1992; McClendon, 1991; Messick, 1991). The convention of incorporating an equal number of positively worded (PW) and reverse worded (RW) items originates from Ray (1983)'s recommendation. In his work, Ray (1983) highlighted the impact of acquiescence bias (ACQ)—the tendency of individuals to agree with statements regardless of their content. He argued that unbalanced or one-way-worded scales could yield inaccurate scores, leading to erroneous relationships among constructs (Ray, 1983). Therefore, researchers have argued that introducing RW items and balancing them with PW items helps mitigate the influence of ACQ bias (e.g., Barnette, 2000; Berkowitz & Wolkon, 1964; Bolt et al., 2014). This is because ACQ bias is canceled out when item scores are aggregated, ensuring that the participant's total score remains unaffected and thus provides a more accurate representation of their true opinions or feelings (e.g., Barnette, 2000; Berkowitz & Wolkon, 1964; Bolt et al., 2014).

Reverse Worded Items and Method Effects

However, researchers have challenged the effectiveness of including RW items and have criticized several negative effects of using RW items (e.g., Lindwall et al., 2012; Savalei & Falk, 2014; Van Sonderen et al., 2013). For example, ACQ bias is argued to be a distinct factor, separate from the substantive factor of the scale, varying across participants and distorting the covariance structure of the items (e.g., Dueber et al., 2021; Savalei &

Falk, 2014; Strizhakova et al., 2008). Consequently, the mere inclusion of RW items is not effective in canceling out the bias (Savalei & Falk, 2014). Furthermore, researchers have demonstrated that RW items typically have higher difficulty and lower item discrimination compared to PW items, which may result in confusion and careless responding (Dueber et al., 2021). These issues associated with RW items primarily arise from method effects due to the wording differences on Likert scales (e.g., DiStefano & Motl, 2006; Lindwall et al., 2012; Marsh, 1996). The method effects, which occur when responses are influenced by factors other than the target construct, introduce systematic variance among some items (e.g., DiStefano & Motl, 2006; Lindwall et al., 2012; Marsh, 1996). Specifically, as highlighted in studies by Zhang and Savalei (2016) and Dueber et al. (2021), the difference in wording between PW and RW items can lead to the emergence of unintended method factors, complicating the interpretation of the substantive factor of the scale and leading to multidimensionality in scales that are theoretically unidimensional (e.g., Carmines & Zeller, 1979; Greenberger et al., 2003; Schmitt & Stuits, 1985).

Over the past few decades, many theoretically unidimensional psychological scales using the Likert format with RW items have been heavily debated regarding the potential effects of these RW items (e.g., Ebesutani et al., 2012; Hensley & Roberts, 1976; Zhang et al., 2016). One of the most well-known scales is the Rosenberg Self-Esteem Scale (RSES), a balanced Likert scale that consists of five PW items and five RW items (Rosenberg, 1989). Theoretically, the RSES is intended to measure global Self-Esteem (GSE) in a unidimensional manner (Rosenberg, 1989). However, a large body of literature have shown that the one-dimensional Confirmatory factory analysis (CFA) models tend to have a poor model fit with data (e.g. DiStefano & Motl, 2006; Gnambs et al., 2018; Greenberger et al., 2003; Lindwall et al., 2012; Marsh, 1996; Marsh et al., 2010). Furthermore, exploratory factor analyses (EFAs) often reveal a two-factor structure in the RSES, with all PW items loading on one factor and all RW items on another (e.g., Greenberger et al., 2003; Hensley & Roberts, 1976; Zhang et al., 2016). A possible

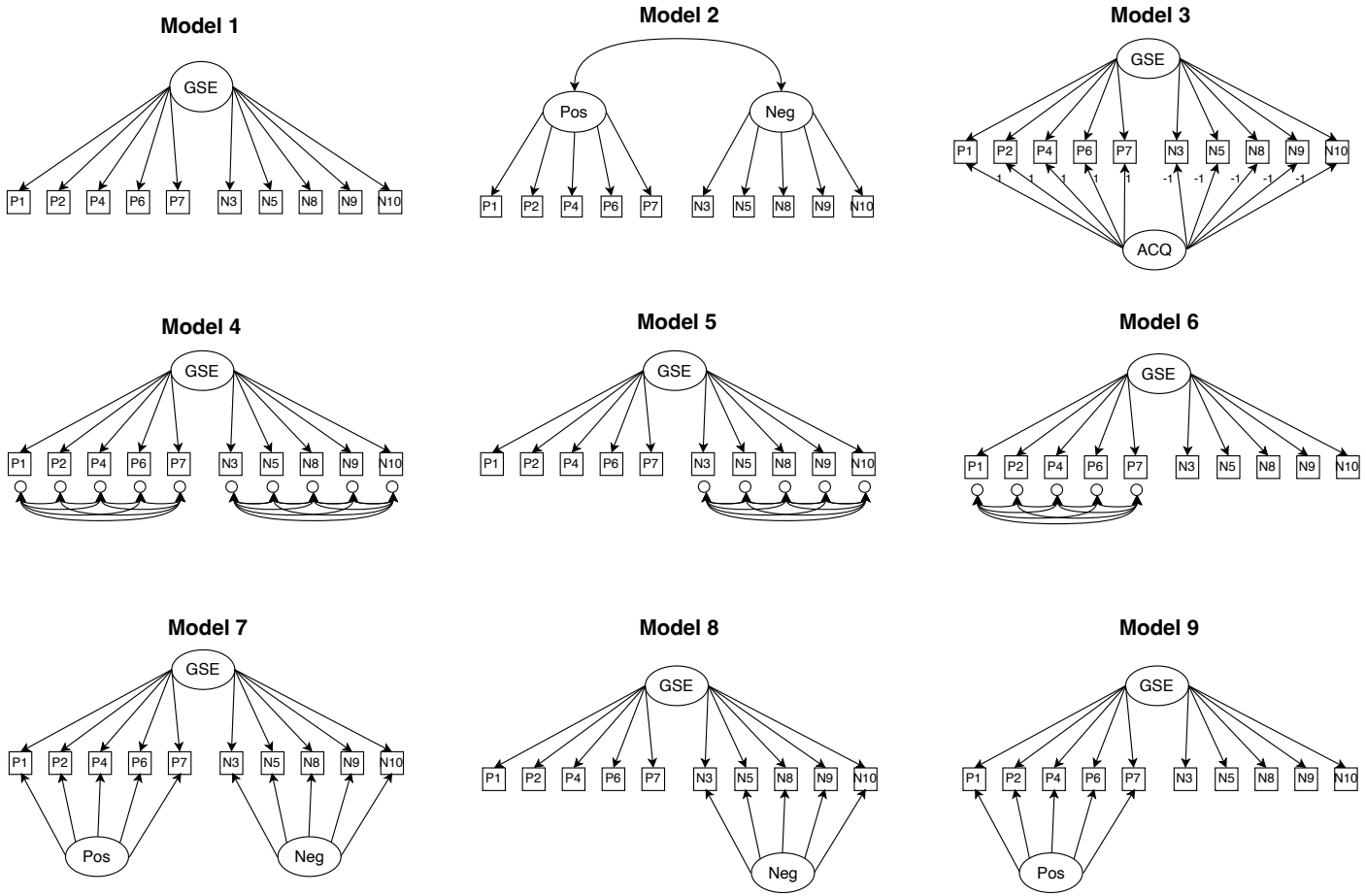
explanation for the two-factor structure is due to the method effects caused by item wording discrepancies (e.g., Greenberger et al., 2003; Hensley & Roberts, 1976; Lindwall et al., 2012). Similar results have been found in other Likert scale formats as well, such as Need for Cognition (e.g., Zhang et al., 2016), Loneliness Questionnaire (e.g., Ebesutani et al., 2012), and Perceived Stress Scale (e.g., Cole et al., 2019). Indeed, such method factors in Likert scales are usually thought to be caused by the introduction of RW items (e.g., Schmitt & Stuits, 1985; Woods, 2006).

CFA Models for Removing Method Effects

To account for the method effects of RW items, researchers have developed various confirmatory factor analysis (CFA) models from the original theoretical one-factor model (e.g., Lindwall et al., 2012; Marsh, 1996; Savalei & Falk, 2014). This original model, which is exemplified in Model 1 in Figure 1 using RSES, posits a single substantive factor without assuming the presence of ACQ bias or method effects (e.g., Lindwall et al., 2012; Marsh, 1996; Marsh et al., 2010). Broadly, there are two primary approaches to constructing these CFA models: the first approach involves explicitly modeling the ACQ bias, while the second focuses on incorporating method factors (e.g., Lindwall et al., 2012; Marsh, 1996; Savalei & Falk, 2014).

Figure 1

CFA Models for Controlling Method effects



Note: Model 1 is the original one-factor model; Model 2 is the two-factor model often appeared in EFA; Model 3 explicitly models ACQ bias; Models 4,5,6 are CTCU-PW/RW, CTCU-RW, CTCU-PW, respectively; Models 7,8,9 are CTCM-PW/RW, CTCM-RW, CTCM-PW, respectively.

To model ACQ bias, researchers have developed a bidimensional model that includes one substantive factor (e.g., GSE) and one ACQ factor, both connected to all PW and RW items (e.g., Billiet & McClendon, 2000; Maydeu-Olivares & Coffman, 2006; Savalei & Falk, 2014). This model distinguishes substantive loadings from ACQ bias by considering the latter as an independent factor and allows unrestricted loadings on scale items (e.g., Billiet & McClendon, 2000; Maydeu-Olivares & Coffman, 2006; Savalei & Falk, 2014). For a detailed illustration, see Model 3 in Figure 1. To ensure model identifiability, loadings on the ACQ factor are set to positive one (on PW items) and negative one (on RW items), with the factor's variance being freely estimated (Savalei & Falk, 2014). Refer to Model 3 in Figure 1 for a visual representation.

On the other hand, multiple models have been suggested in order to directly model the method effect (e.g., DiStefano & Motl, 2006; Lindwall et al., 2012; Marsh, 1996). Some researchers have proposed the model with two correlated method factors, as illustrated in Model 2 in Figure 1 using RSES—one connects on PW items, the other connects on RW items, while omitting the substantive factor (e.g., Kaplan & Pokorny, 1969; Owens, 1993; Tafarodi & Swann Jr, 1995). However, one major issue of this model is that the simplistic two-correlated-factor model does not allow researchers to clearly distinguish between the substantive factor (e.g., GSE) and the method factors (Carmines & Zeller, 1979; Owens, 1993). In response to this, more models have been proposed and widely adopted to distinguish the substantive factor and method factors based on the multitrait-multimethod (MTMM) conceptual framework (e.g., Bagozzi et al., 2003; Lindwall et al., 2012; Marsh, 1996). These models fall into two primary categories: the Correlated Trait, Correlated Uniqueness (CTCU) models and the Correlated Trait, Correlated Methods (CTCM) models. The CTCU models represent method effects by including correlations among the residuals or uniqueness on the PW items (CTCU-PW model; see Model 6 in Figure 1) or RW items(CTCU-RW Model; see Model 5 in Figure 1) or both (CTCU-PW/RW Model; see Model 4 in Figure 1), assuming that method effects are ephemeral artifacts due to

wording effects that have no substantive relevance (e.g., Bagozzi et al., 2003; Lindwall et al., 2012; Marsh, 1996). In contrast, the CTCM models introduce specific method effect factors connecting all PW items (CTCM-PW Model; see Model 9 in Figure 1) or all RW items (CTCM-RW Model; see Model 8 in Figure 1), or both (CTCM-PW/RW model; see Model 6 in Figure 1), assuming the method effects represent stable response styles (e.g., Bagozzi et al., 2003; Lindwall et al., 2012; Marsh, 1996). In this paper, we will call CTCU and CTCM models that model only PW or RW items (e.g., CTCU-RW, CTCU-PW, CTCM-RW, CTCM-PW) "unidirectional models", and those that are modeling both PW and RW items (e.g., CTCU-PW/RW, CTCM-PM/RW) "bidirectional models".

To find out which model is the best and should be adopted, researchers have compared the fit indices among these CFA models (e.g., Lindwall et al., 2012; Marsh, 1996; Marsh et al., 2010). The original one-factor model (Model 1) was not able to adequately fit the data, making it the worst-fitting model (e.g., DiStefano & Motl, 2006; Lindwall et al., 2012; Marsh et al., 2010). The two-correlated-factors model (Model 2) showed an improved fit over Model 1, but was still outperformed by the MTMM models (Models 4-9) (e.g., DiStefano & Motl, 2006; Lindwall et al., 2012; Marsh et al., 2010). Within the MTMM models, models that incorporate both PW and RW method effects (CTCU Model 4 and CTCM Model 7)generally exhibited better fits than those with unidirectional method effects (CTCU Models 5, 6 and CTCM models 8, 9). However, comparisons between the CTCU and CTCM models incorporating both PW and RW effects yielded inconsistent results across studies (e.g., Lindwall et al., 2012; Marsh et al., 2010). For instance, while Marsh et al. (2010) found Model 7 to have the best fit, suggesting stable responding styles, Lindwall et al. (2012) reported a superior fit for Model 4. For CTCU or CTCM models that incorporate only unidirectional method effect (CTCU Models 5 and 6, CTCM Models 8 and 9), the models that included only PW method effects have similar fits to those models that include only RW method effects (e.g., Carmines & Zeller, 1979; Lindwall et al., 2012; Marsh et al., 2010). Similar to the MTMM models, the bidimensional model with the

ACQ factor (Model 3) significantly improved the fit compared to the original one-factor model (e.g., Billiet & McClendon, 2000; Maydeu-Olivares & Coffman, 2006; Savalei & Falk, 2014). However, the comparisons between this model and the MTMM models (CTCM and CTCU) are rare, leaving the relationship among these models less clear and highlighting the need for more extensive comparative analysis.

Nesting Relationship of CFA Models

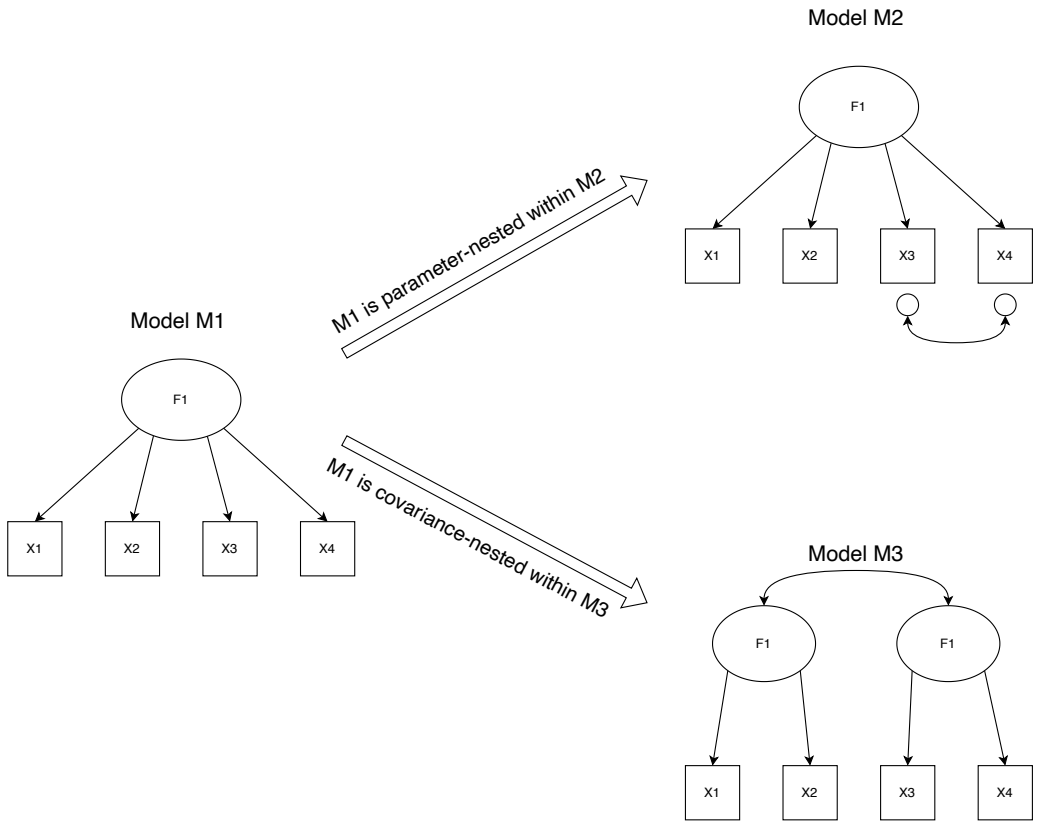
Indeed, many studies have shown that the alternative CFA models had a better fit than the theoretical model of the scale (i.e., the original 1-factor model) because they may be better reflective of the underlying covariance structure influenced by method effects (Greenberger et al., 2003; Hensley & Roberts, 1976; Zhang et al., 2016). However, this improvement in model fit could also result from the nesting structure among the CFA models in Figure 1. The nesting relationship between any two CFA Models, A and B, can be defined as follows: Model A is nested within Model B if the set of all covariance matrices possible under Model A is a subset of the set of all covariance matrices possible under Model B (e.g., Asparouhov & Muthén, 2019; Bentler, 1990; Bentler & Satorra, 2010). This means Model A is more restrictive and more parsimonious than Model B (e.g., Asparouhov & Muthén, 2019; Bentler, 1990; Bentler & Satorra, 2010). Moreover, if Model A is nested within Model B, the less restrictive Model B will have a better fit than Model A (Asparouhov & Muthén, 2019).

Researchers have identified two types of nesting based on the definition: 1) Parameter nesting; 2) Covariance nesting (e.g., Asparouhov & Muthén, 2019; Bentler, 1990; Bentler & Satorra, 2010). Parameter nesting is the more understood type in which the more restricted model can be obtained by a) fixing some parameter or b) imposing constraints from the more complicated model (Bentler & Bonett, 1980). An example is seen in Figure 2, where Model M1 is parameter-nested within Model M2 because one can obtain the more restricted Model M1 by constraining the correlation between the residuals of the indicators X3 and X4. This also implies that Model M2 will have a better model fit

than Model M1. Similarly, within the aforementioned CFA models, the one-factor model (Model 1) is nested within CTCU-RW (Model 5) in the parameter sense because it can be obtained by imposing constraints among the RW item residuals in Model 5. While many parameter nesting examples can verified visually, reliance on computerized tools is advisable to minimize human error (Bentler & Satorra, 2010).

Figure 2

Nesting Structure Examples



Note: Model M1 is parameter nested within M2, and covariance-nested within M3.

Covariance nesting, on the other hand, is less straightforward to verify visually, as it does not involve the fixing of parameter(s) or imposing constraint(s) like parameter nesting and yet still meets the requirement for nesting (Asparouhov & Muthén, 2019; Bentler & Satorra, 2010). For example, although Models M1 and M3 in Figure 2 may seem quite different, Model M1 is actually covariance-nested within Model M3. Similarly, we suspect

that there might be some covariance nesting structures among the CFA models that are not immediately apparent, which could also contribute to the improvement in model fit beyond the explanation that some models are better at modeling method effects.

Present Study

The goal of the study is to examine the nesting structures of all the models in Figure 1. We will use a method called the nesting and equivalence testing (NET) procedure (Bentler & Satorra, 2010). The NET procedure is a three-step method developed by Bentler and Satorra (2010) that allows researchers to determine whether two models are nested using a real dataset. As highlighted in their paper, the NET procedure is able to test the nesting relationship regardless of whether it is parameter or covariance nesting (Bentler & Satorra, 2010). The rest of the paper will be organized as follows. In the next section, we will explain the NET procedure in detail and elaborate on how we can test the nesting structure of the models in Figure 2 using the NET procedure. Then, we will use two empirical Likert scale datasets to compare the model fit among all nested models. These comparisons will be done using a fit index called RMSEA_D, which will be explained later. By comparing the model fit among the nested models, we can determine which models better represent the structure of the Likert scales. We end the papers by discussing the implications of our results and offer recommendations to applied researchers.

Testing the Nesting Structure Using NET Procedure

NET Procedure

The Nesting and Equivalence Testing (NET) procedure, as developed in Bentler (2010), provides a systematic approach for examining the nesting relationships between SEM (e.g., CFA) models. This method assesses whether one CFA model is nested within another using their respective covariance matrices. The following steps outline the NET procedure:

Step 1. Conduct a SEM analysis on the first CFA model, M_1, using the observed covariance

matrix S. Record the degrees of freedom (df_1) and the model-reproduced covariances $\widehat{\Sigma}_1$.

Step 2. Use $\widehat{\Sigma}_1$ as the input covariance matrix for a subsequent SEM analysis on the second CFA model, M_2. Save the resulting degrees of freedom (df_2) and the chi-square statistic ($\widehat{F}$).

Step 3. Calculate the difference in degrees of freedom, $df_D = df_1 - df_2$, and set a small criterion value for $\widehat{F}$ (e.g., $\varepsilon = .00001$).

Step 4. Interpret the results as follows:

(a) If $df_D > 0$ and $\widehat{F} < \varepsilon$, M_1 is nested within M_2.

(b) If $df_D = 0$ and $\widehat{F} < \varepsilon$, M_1 and M_2 are equivalent.

(c) If $df_D < 0$ or $\widehat{F} \geqslant \varepsilon$, M_1 is neither nested within nor equivalent to M_2.

This method extends beyond visual comparisons, making it able to work precisely for assessing if the nesting relationship holds for two models regardless of whether they are parameter-nested or covariance-nested (e.g., Asparouhov & Muthén, 2019; Bentler & Satorra, 2010). If $\widehat{F} < \varepsilon$, it indicates that M_2 can exactly reproduce the covariances of M_1, suggesting that the possible covariance matrices under M_1 are a subset of those under M_2. This aligns with the definition of model nesting (Bentler & Satorra, 2010). Conversely, if $\widehat{F} \geqslant \varepsilon$, the models are not nested or equivalent. The criterion ε reflects computational precision and may be adjusted as computational capabilities evolve (Satorra, 1989; Steiger et al., 1985). However, one uncertainty associated with NET procedure is the local nature of NET procedure, meaning that NET results cannot 100% guarantee that the nesting structure holds for all possible datasets; however, most of the time, NET results hold in all possible datasets (see Bentler & Satorra, 2010, for more details)

For this study, we will use the NET procedure to investigate the nesting relationship among the nine CFA models in Figure 1. Specifically, we will conduct a model nesting

comparison for each possible pair of models (i.e., treating one model as M_1 (the model with the larger df) and the other as M_2 (the model with the fewer df) for each of the 36 possible pairs of models). The dataset providing the observed covariance matrix S for the original SEM runs are well-established surveys that incorporate a Likert scale as well as a balanced number of PW and RW items: Rosenberg Self-Esteem Scale (RSES) and Need for Cognition Scale (NFC) (Zhang et al., 2016, 2023).

Results of NET Procedure

After conducting the nesting relationship comparison for each possible pair using the NET procedure, we were able to uncover the full nesting relationship among the 9 CFA models in Figure 1 as presented in Table 1.

Table 1
Nesting Structure of the CFA Models in Figure 1

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Model 1	1 in 1	1 in 2	1 in 3	1 in 4	1 in 5	1 in 6	1 in 7	1 in 8	1 in 9
Model 2		2 in 2	NN	2 in 4	2 in 5	2 in 6	2 in 7	2 in 8	2 in 9
Model 3			3 in 3	NN	NN	NN	NN	NN	NN
Model 4				4 in 4	5 in 4	6 in 4	7 in 4	8 in 4	9 in 4
Model 5					5 in 5	NN	NN	8 in 5	NN
Model 6						6 in 6	NN	NN	9 in 6
Model 7							7 in 7	8 in 7	9 in 7
Model 8								8 in 8	NN
Model 9									9 in 9

Note: "1 in 2" means that Model 1 is nested within Model 2. NN=not nested.

For the nesting relationship involving Model 1 (original 1-factor model), we observed that Model 1 is nested within all other CFA models as shown in Figure 1. Specifically, Model 1 is parameter-nested within the CTCU models (Models 4-6), the CTCM models (Models 7-9), and the ACQ model (Model 3). This is because we can obtain Model 1 by constraining all correlations between the uniquenesses of items for CTCU models, by constraining all regressions with the method factor for CTCM models, and by constraining all regressions with the ACQ factor for the ACQ model. Additionally, Model 1 is covariance-nested within the two-correlated-factor model (Model 2), as we

cannot visually confirm its parameter nesting.

For the nesting relationship involving Model 2 (two-related-factor model), we observed that Model 2 is covariance-nested within all CTCU and CTCM models, as we cannot obtain Model 2 by constraining any paths from CTCU or CTCM models. However, Model 2 is not nested within Model 1 nor Model 3, since both Model 1 and Model 3 have more degrees of freedom than Model 2.

Regarding the nesting relationship involving Model 3 (ACQ factor model), we observed that Model 3 is not nested within any of the CTCU nor CTCM models, although Model 3 has more degrees of freedom than any of them.

For the nesting relationships involving CTCU (Models 4-6) and CTCM (Models 7-9), we observed a pattern where: 1) the unidirectional models are nested within the bidirectional models within and between CTCM and CTCU frameworks 2) for models with the same direction or with both directions, those with method factors are nested within those with correlated uniqueness; namely, the CTCM models are nested with their CTCU counterparts respectively, and 3) the unidirectional models within each CTCU or CTCM framework are not equivalent (i.e., not nesting with the same degrees of freedom) to each other. For point 1), we observed that the unidirectional models CTCU-RW (Model 5) and CTCU-PW (Model 6) are nested within the bidirectional model CTCU-PW/RW (Model 4) within the CTCU framework. Similarly, the unidirectional models CTCM-RW (Model 8) and CTCU-PW (Model 9) are nested within the bidirectional model CTCM-PW/RW (Model 7) within the CTCM framework, as assessed visually through parameter nesting. Furthermore, between the two frameworks, we observed that the unidirectional models in CTCM(CTCM-RW and CTCM-PW) are nested within the bidirectional model CTCU-PW/RW in the CTCU framework. Regarding point 2), we observed that the all CTCM models CTCM-RW (Model 8), CTCM-PW (Model 9), and CTCM-PW/RW(Model 7) are nested within their counterparts in the CTCU framework, CTCU-RW (Model 5), CTCU-PW (Model 6), and CTCU-PW/RW (Model 4), respectively. For point 3), we

observed that although having the same degrees of freedom and appearing symmetric with each other, unidirectional models within the same framework (CTCU or CTCM) are not equivalent to each other; that is, there is no nesting relationship between CTCU-PW and CTCU-RW, nor between CTCM-PW and CTCM-RW.

Comparing Model Fit on Empirical Data Using $\mathbf{RMSEA_D}$

In the previous section, we figured out the nesting structures among all the CFA models for Likert scales shown in Figure 1. In this section, we will compare the model fit between each pair of nested models using Likert scale datasets found in published psychological studies, namely, Rosenberg Self-Esteem Scale (RSES) and Need for Cognition (NFC) scale. To compare the model fit between two nested models for each dataset, we will use a difference test based on the value of the root mean square error approximation (RMSEA), which is called RMSEA_D, following Savalei et al. (2021). Let N be the sample size, T_1 and df_1 be the chi-square statistic and the degree of freedom for the more restricted model and T_2 and df_2 be the ones for the less restricted model. Then RMSEA_D is defined as follows:

$$\begin{aligned} \text{RMSEA}_D &= \sqrt{\frac{(T_1 - T_2) - (df_1 - df_2)}{(df_1 - df_2)(N-1)}} \\ &= \sqrt{\frac{D - df_D}{df_D(N-1)}}, \end{aligned} \tag{1}$$

where $D = T_1 - T_2$ is the difference in the chi-square statistics between the more restrictive model and the less restrictive model and $df_D = df_1 - df_2$ is the difference in the degrees of freedom between the more restrictive model and the less restrictive model. RMSEA_D is considered the most direct test of the close fit of the proposed constraints as it does not depend on df_A and thus does not lose sensitivity in larger models (Savalei et al., 2021). Furthermore, RMSEA_D is interpreted in the same way as the overall model RMSEA as the amount of additional misfit introduced by the constraints in the more restricted model per additional degree of freedom gained; therefore, a value greater than 0.08 indicates a

substantial increase in misfit due to the constraints introduced by the more restricted model (Savalei et al., 2021).

Likert Scales

Rosenberg Self-Esteem Scale

Rosenberg Self-Esteem Scale (RSES), as introduced by Rosenberg (1989), was designed to measure global self-esteem (GSE). This scale is a 10-item Lickert scale that contains five PW items and five RW items (Rosenberg, 1989). Each item is measured on a four-point scale, with 4 corresponding to *Strongly agree* (high in GSE) and 1 corresponding to *Strongly disagree* (low in GSE), and participants are asked to select one of the four options that best describes them (see Appendix). The dataset we used for representing RSES in this study is the original version of RSES from Zhang et al. (2019), in which they created different versions of RSES (including the original version) and compared the factor structure of different versions.

Need For Cognition Scale

Need for Cognition (NFC) scale was developed to measure Need for Cognition (NFC), which is theorized as a stable personality trait reflecting the person's intrinsic motivation to engage in and enjoy cognitive activities (Cacioppo et al., 1983). NFC scale contains 18 items with nine PW items and nine RW items. Each item is measured on a nine-point scale, with 9 corresponding to *very strong disagreement* (low in NFC) and 1 corresponding to *very strong agreement* (high in NFC) The dataset we used for representing NFC scale in this study is from Zhang et al. (2016), in which they compared the factor structure after manipulating the number and type of RW items.

Results of $\mathbf{RMSEA_D}$ Analyses

Tables 2 and 3 show the RMSEA_D analyses using data from the RSES and NFC scale. Across the two Likert scales, there are several consistent patterns of results. First, Model 1 had a substantial increase in misfit per additional degree of freedom compared

with the rest of the models, meaning that Model 1 is substantially worse-fitting than the other models. Similarly, Model 2 also had a substantial increase in misfit per additional degree of freedom compared with all the MTMM Models (i.e., Models 4-9), meaning that Model 2 is worse-fitting than the MTMM models. Furthermore, comparing CTCU models with CTCM models, $RMSEA_D$ results show that most CTCU models have substantial improvement in fit compared with the corresponding CTCM models with the CTCU-PW/RW model appears to have better fit on all other MTMM models except Model 7 (CTCM-PW/RW). Specifically, the $RMSEA_D$ values between Model 4 and Model 7 in both datasets are one of the lowest $RMSEA_D$ (0.004 for RSES and 0.077 for NFC), meaning that the misfit introduced by Model 4 compared to Model 7 is trivial. This pattern of results shows although the CTCU-PW/RW model is structurally more complex than all other MTMM models and often has better model fit as indicated by chi-square statistic and other fit indices, relative to this complexity, the CTCU-PW/RW model's fit is equivalent to the CTCM-PW/RW model's fit; in this case, the simpler model (CTCM-PW/RW) should be preferable. Moreover, within CTCM and CTCU Models, we observed that the bidirectional model CTCM-PW/RW has a substantial increase in model fit compared with unidirectional models CTCM-RW (with $RMSEA_D$ of 0.164 for RSES and 0.141 for NFC scale) and CTCM-PW (with $RMSEA_D$ of 0.244 for RSES and 0.087 for NFC scale), and the bidirectional model CTCU-PW/RW has a substantial increase in model fit compared with unidirectional models CTCU-RW (with $RMSEA_D$ of 0.119 for RSES and 0.076 for NFC scale) and CTCU-PW (with $RMSEA_D$ of 0.138 for RSES and 0.092 for NFC scale).

In addition, some patterns of results are different across the two Likert scales. For example, for the RSES, the $RMSEA_D$ comparing the CTCM-RW and CTCU-RW models (i.e., Models 5 and 8) is 0, much smaller than the $RMSEA_D$ comparing the CTCM-PW and CTCU-PW models (i.e., Models 6 and 9), which is 0.146. However, for the NFC scale, the $RMSEA_D$ comparing Models 5 and 8 (with $RMSEA_D$ of 0.104) is greater than that comparing Models 6 and 9 (with $RMSEA_D$ of 0.055).

Table 2

$RMSEA_D$ Using NFC Scale Based on the Nesting Results in Table 1

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Model 1	0	1.215	1.230	0.167	0.222	0.216	0.304	0.407	0.421
Model 2		0	NN	0.085	0.093	0.076	0.106	0.038	0.123
Model 3			0	NN	NN	NN	NN	NN	NN
Model 4				0	0.076	0.092	0.077	0.089	0.079
Model 5					0	NN	NN	0.104	NN
Model 6						0	NN	NN	0.055
Model 7							0	0.141	0.087
Model 8								0	NN
Model 9									0

Note: In the diagonal the $RMSEA_D$ is 0 for each model since each model is equivalent to itself.

Table 3

$RMSEA_D$ Using RSES Scale Based on the Nesting Results in Table 1

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Model 1	0	0.372	0.383	0.145	0.177	0.164	0.206	0.248	0.194
Model 2		0	NN	0.122	0.139	0.120	0.178	0.205	0.112
Model 3			0	NN	NN	NN	NN	NN	NN
Model 4				0	0.105	0.124	0.004	0.088	0.125
Model 5					0	NN	NN	0.032	NN
Model 6						0	NN	NN	0.126
Model 7							0	0.152	0.216
Model 8								0	NN
Model 9									0

Note: In the diagonal the $RMSEA_D$ is 0 for each model since each model is equivalent to itself.

Discussion

The inclusion of RW items has been shown to distort the covariance factor structure of the Likert Scale despite its positive effect in canceling out the effects of acquiescence bias (e.g., Asparouhov & Muthén, 2019; Barnette, 2000; Berkowitz & Wolkon, 1964). As a result, multiple new factor models have been proposed to better model the acquiescence bias and the effects of RW items (e.g., Lindwall et al., 2012; Marsh, 1996; Savalei & Falk, 2014). The series of models includes the original one-factor model without modeling method effects or ACQ bias (i.e., Model 1 in 1), the two-factor model that often appears from EFA (i.e., Model 2 in 1), the model that explicitly models ACQ bias (i.e., Model 3 in 1), CTCU Models (CTCU-PW/RW, CTCU-RW, CTCU-PW) corresponding to Models 4,

5, and 6 in 1, and CTCM Models (CTCM-PW/RW, CTCM-RW, CTCM-PW) corresponding to Models 7, 8, and 9 in 1. The present study aims to find the preferable model for Likert scales with RW items by determining their nesting relationships and applying a model comparison test with $RMSEA_D$ fit indices.

We first investigated the nesting structure among these models using the NET procedure (Bentler & Satorra, 2010). The results show that 1) Model 1 is nested within all other CFA models, 2) Model 2 is nested within all other MTMM models (CTCU and CTCM models), and 3) Model 3 is not nested within any other models. Regarding the MTMM models, we discovered that models that only model one direction (PW or RW) are nested within models that address both directions (PW and RW) within each CTCU or CTCM branch, and CTCM models are nested within their counterparts in the CTCU framework (i.e., CTCM-PW/RW in CTCU-PW/RW, CTCM-RW in CTCU-RW, CTCM-PW in CTCU-PW).

After determining the nesting relationship between the CFA models, we used two balanced Likert scales (i.e., NFC Scale and RSES) to compare the model fit of the nested models via the fit indices $RMSEA_D$ (Savalei et al., 2023). Several noticeable patterns emerged in our $RMSEA_D$ results. First, the original one-factor models appear to have a much lower model fit than other CFA models, successfully replicating the findings of existing studies but with a different method and indices (e.g., Lindwall et al., 2012; Savalei et al., 2023). Second, we observed that models accounting for method effects affecting both PW and RW items fit better than those addressing only one orientation. This suggests that the impact of RW items on the factor structure increases the covariance among not only the RW items but also the original PW items. Finally, the fact that the simpler CTCU-PW/RW model has a similar fit to the more complex CTCM-PW/RW model suggests that modeling the method effects as a separate factor or as correlated uniqueness has a similar effect on restoring the factor structure, leading us to conclude that the CTCM-PW/RW is the most effective in modeling the factor structure because it is more

parsimonious while being well-fitted. Furthermore, this might also provide a plausible reason for the inconsistent results when comparing CTCU and CTCM models that only model one direction (PW or RW),

Limitations

Despite the novel approach of this study in determining the most suitable factor structure representation for Likert scales with RW items, there are several limitations worth considering. First, all models or factor structures proposed and assessed in this study are based on balanced scales (i.e., an equal number of PW and RW items), making the conclusions only applicable to Likert scales with an equal number of PW and RW items. Consequently, our findings may not be generalized to other Likert scales. More specifically, many Likert scales are unbalanced, meaning they have an unequal number of PW and RW items. This imbalance between PW and RW items may change the nesting structures among the models, consequently affecting the differences in the fit of the models. Future research should, therefore, incorporate a variety of Likert scales, extending beyond balanced scales to include unbalanced scales, which might reveal different patterns of influence by method effects compared to balanced scales.

Second, the absence of a nesting relationship with Model 3 has restricted it from being considered in the analysis of model fit, thus preventing an assessment of whether modeling the ACQ bias would generate a better fit than the CTCM-PW/RW model. Future studies might also consider using procedures that can compare non-nested models, such as the Likelihood ratio test (LRT) method introduced in (Merkle et al., 2016) for testing non-nested structural equation models.

Conclusion

In conclusion, our study systematically examined the nesting structures among various confirmatory factor analysis (CFA) models designed to account for method effects and acquiescence bias in Likert scales. Through a series of comparisons using RMSEAD, we determined that the CTCM-PW/RW model is the optimal measurement model for

balanced Likert scales. This study contributes to resolving the inconsistent results concerning the effectiveness of different CFA models through a more rigorous process. Therefore, we recommend that researchers use the CTCM-PW/RW model as the measurement model for balanced Likert scales, especially when including measurement models of Likert scales in structural equation modeling analyses.

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